

Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)—applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

subsectionResearch Aim and Questions How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021, Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq 90\%$) with low false alarm rates — acceptable FARs were 1-2/week for patients with ≥ 1 seizure/week and 1-2/month for patients with < 1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. Therefore, the design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

Simultaneously, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

Consequently, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most robust approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE’s phased framework (Phases 0–4) (Beniczky et al., 2021). Retrospective studies (Phases 0-2), while foundational, risk significant data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025). The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The final step is in-field evaluation (Phase 4) in patients’ homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. Search strategy (Multi-stage, 2015–2025):

- *Scoping*: Google Scholar for broad coverage and identifying candidate papers.
- *Formal queries*: IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
- *Citation chasing*: Backward and forward citation tracking on key papers to ensure completeness.

2. Inclusion / Exclusion criteria:

- *Inclusion*: Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
- *Exclusion*: EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.

3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
 - *Keywords:* ‘epilepsy’, ‘seizure’, ‘ictal’, ‘preictal’
 - *MeSH Terms (PubMed):* Epilepsy, Seizures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
 - *Biosignal Keywords:* ‘electrocardiogram’, ‘ECG’, ‘heart rate variability’, ‘HRV’, ‘photoplethysmography’, ‘PPG’, ‘electrodermal activity’, ‘EDA’, ‘accelerometer’, ‘ACC’
 - *Technology Keywords:* ‘wearable’, ‘wearable device’
 - *Method Keywords:* ‘deep learning’, ‘machine learning’, ‘convolutional neural network’, ‘CNN’, ‘long short-term memory’, ‘LSTM’, ‘neural network’
 - *MeSH Terms (PubMed):* Wearable Electronic Devices, Electrocardiography, Deep Learning

3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
 - *Exclusion Keywords:* ‘EEG’, ‘electroencephalography’, ‘iEEG’, ‘intracranial’
 - *Search Logic:* The NOT operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
 - *Keywords:* ‘detection’, ‘prediction’, ‘sensitivity’, ‘specificity’, ‘false alarm rate’, ‘FAR’, ‘area under the curve’, ‘AUC’

/subsectionExploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were acquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

4.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015–2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter *EEG-only modalities*, not to exclude studies that mention EEG solely as a reference standard.

5 Study Comparison Following Webster and Watson

Following the concept matrix approach proposed by (Webster & Watson, 2002), this section presents a structured comparison of the 13 identified studies on wearable seizure detection and forecasting. The matrix organizes studies along key dimensions relevant to clinical translation and technical implementation.

5.1 Concept Matrix: Study Overview

Table 1 provides a comprehensive comparison of all included studies, organized by primary objective, methodology, and outcomes. The matrix enables cross-study analysis of trends in sensor modalities, algorithmic approaches, and validation strategies across the 2013–2026 period.

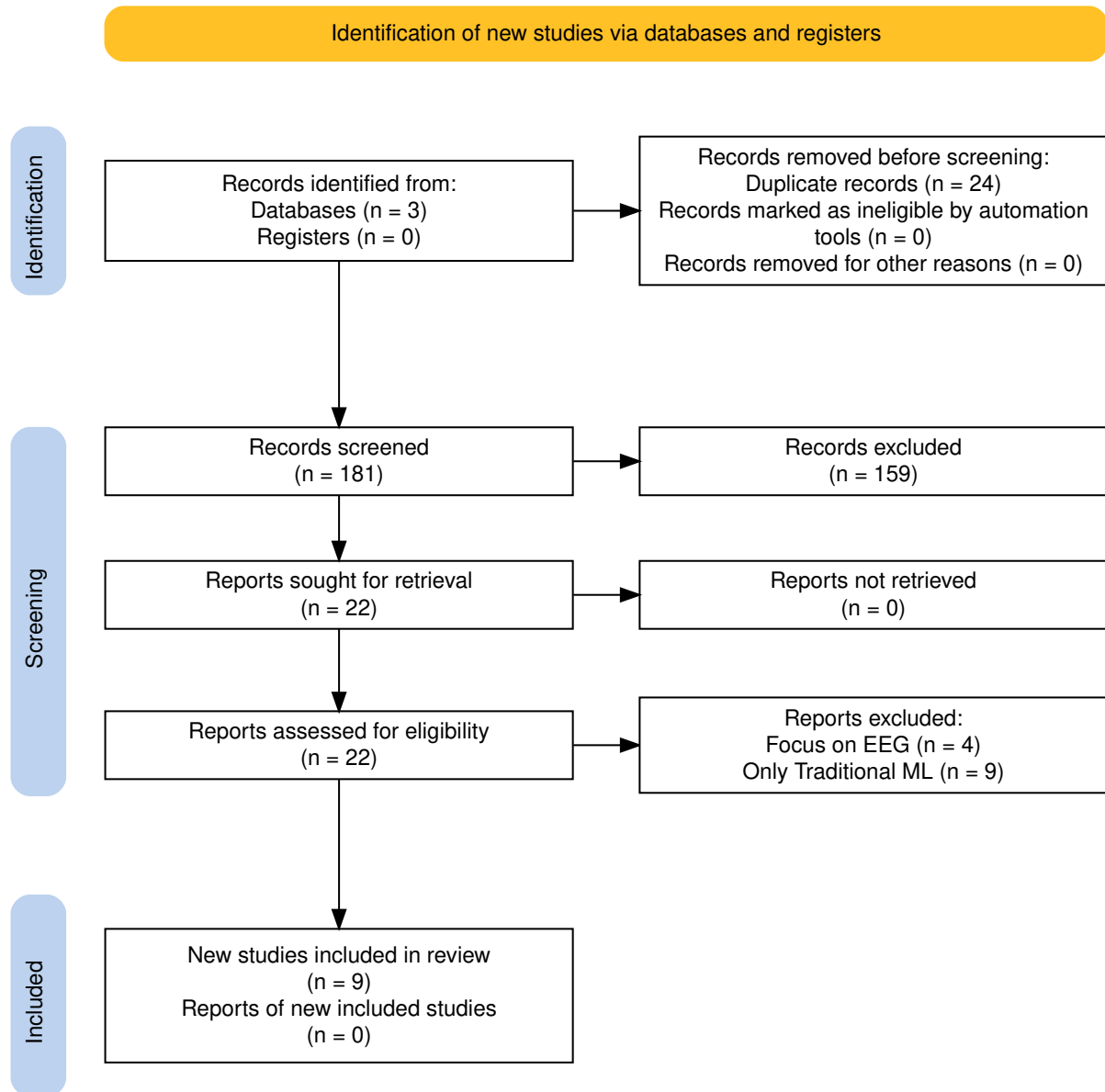


Figure 1: PRISMA flow diagram of the study selection process.

Table 1: Webster & Watson Concept Matrix: Wearable Seizure Detection and Forecasting Studies (2013–2026)

Study	Objective	Design	Sample	Device	Loc.	Algorithm	Sens	Spec	FPR	Key Limitation
Detection Studies										
Spahr et al. 2025	Convulsive multi-center DET	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ensemble 1D CNN	96%	–	<1/8d	EMU, offline
Reintjes et al. 2025	ECG anomaly DET	Retro	n=120, 856 szs	Single-lead ECG	Chest	Matrix Profile	98.16%	–	1.91/h	High FAR
Fine 2025	Tonic DET	Phase 1	n=15/3	6-axis band	Wrist	ANN (594 feat)	100%	–	0.23/n; 0.023/h	Small test
Dong et al. 2026	Nocturnal severe szs	Prospective	n=68, 1846 szs	NightWatch	Arm	CNN-LSTM	71.6%	–	0.165/h	Single-center
Wang et al. 2025	Sz DET (acc: 97.8%)	Retro	n=28, 62 szs	Biovital-P1	Wrist	LSTM+SVM/LDA	–	–	8.5/24h	Hospital only
Singh Rathore 2024	Multimodal	Single pt	1 pt, 6 hrs	EDA+ACC+HR	–	MLP, SVM, KNN	96.8%	–	–	Single patient
Elemam et al. 2025	Q+HRV+Audio	Cross-sect	Q:198, HRV:30	Camera PPG	Finger	CNN, rule based	95.1%	97.1%	–	Small samples
Borujeny 2013	Wireless ACC	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	KNN k=5	100%	–	0	Very small
Forecasting Studies										
Vieluf et al. 2025	Diary+wearable	Retro	n=70, 5437 d	Embrace	Wrist	DNN+harmonic	82%	67%	–	Daily res only
Meisel et al. 2020	LOSO forecast	LOSO CV	n=69, 452 szs	Empatica E4	Wrist	LSTM+1DConv	51.2%	–	TiW 43.7%	Pediatric
Stirling 2021	HR cycle forecast	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	LSTM+RF+LR	–	–	AUC 0.74	Self-report
Nasseri 2021	Ambulatory forecast	Retro	n=6	Empatica E4	Wrist	LSTM 4-layer	AUC 0.75	–	TiW 0.9–7.2h/d	RNS required
Ode et al. 2023	RRI prediction	Retro	n=66, 85 szs	ECG	–	Self-Att AE	74%	–	0.85/h	FPs in some pts

Note: ACC = accelerometer; DET = detection; ANN = artificial neural network; AE = autoencoder; CNN = convolutional neural network; CS = convulsive seizure; DNN = deep neural network; ECG = electrocardiogram; EDA = electrodermal activity; HR = heart rate; HRV = HR variability; LOSO = leave-one-subject-out; LSTM = long short-term memory; PPG = photoplethysmography; RRI = R-R interval; Sens = sensitivity; Spec = specificity; sz(s) = seizure(s); FPR = false positive rate; FA = false alarm; TiW = time in warning; – = not reported.

5.2 Detailed Clinical and Technical Metrics

Table 2 presents additional clinically relevant metrics that are essential for evaluating real-world applicability but often underreported in the literature. These metrics address key questions about patient-level performance, detection timeliness, and generalizability.

Table 2: Detailed Clinical and Technical Metrics Across Studies

Study	Validation	Detect. Latency	Patient Success	Precision/PPV	Other Metrics	Clinical Notes
Detection Studies						
Spahr et al. 2025	Indep. test (347 pt)	26 s median	–	–	HMS 92.6	Phase 2; 2 CSs missed
Reintjes et al. 2025	Retro (SeizeIT2)	–	Responders only	–	HMS 92.6	FAR $\gg$ clinical target
Fine 2025	Indep. test (3 pt)	14 s mean	–	–	95% CI: 69–100	Phase 1 only
Dong et al. 2026	10-fold CV	<30 s	–	–	AUC 0.793	First long-term home
Wang et al. 2025	Hospital retro	–	7/22 pts	83–87%	Acc 97.8%	Single hospital
Singh Rathore 2024	Single pt	–	–	94.8%	F1 95.6%	3.2M data pts
Elemam et al. 2025	Cross-sect	–	–	90.3%	–	PPG 15 s only
Borujeny 2013	3-pt retro	–	3/3 pts	–	–	20 szs total
Forecasting Studies						
Vieluf et al. 2025	Retro longitudinal	–	–	–	F1 0.81	Daily predictions
Meisel et al. 2020	LOSO CV	–	30/69 (43.5%)	–	IoC 14.1%	Pediatric only
Stirling 2021	Pseudo-prospective	–	100% hourly	–	AUC 0.74	HR cycles key
Nasseri 2021	Retro analysis	33 s mean	5/6 (83%)	–	AUC 0.75 (0.50–0.92)	RNS validation req
Ode et al. 2023	Retro	–	29/66 (44%)	0.35	AUC 0.97	Individual tuning

Note: CV = cross-validation; HMS = Harmonic Mean Score (Sensitivity – 0.4×FAR); IoC = Improvement over Chance; LOSO = leave-one-subject-out; Indep. = independent; Responders = patients with >50 BPM HR change during seizures; HMS = Sensitivity – 0.4×FAR (Beniczky & Ryvlin, 2018); Patient Success = proportion achieving better-than-chance or usable performance.

5.3 Dimensional Analysis

5.3.1 Sensor Modality Trends

- **Wrist-worn accelerometry** dominates (8/13 studies), leveraging commercial devices (Empatica E4, Fitbit, Embrace).
- **Cardiac signals** (ECG, PPG, HRV) show increasing adoption for preictal detection, with 6 studies incorporating autonomic measures.
- **Multi-modal fusion** (ACC + EDA + PPG) appears in 4 recent studies, reflecting recognition that single modalities have limited specificity.

5.3.2 Algorithmic Evolution

- **2013:** Traditional ML (KNN, ANN) with handcrafted features (Borujeny et al., 2013).
- **2020–2021:** Emergence of LSTM and ensemble methods (Meisel et al., 2020; Stirling et al., 2021).
- **2023–2026:** Deep learning dominance with CNNs, attention mechanisms, and anomaly detection approaches (Ode et al., 2023; Reintjes et al., 2025; Spahr et al., 2025).

5.3.3 Validation Rigor

- **Patient-level preservation:** Only 4 studies use LOSO or patient-independent splits.
- **Prospective validation:** Only 3 studies (Dong et al., 2022; Elemam et al., 2025; Spahr et al., 2025) include prospective phases.
- **Real-world testing:** Limited to 2 studies (Nasseri et al., 2021; Vieluf et al., 2025); most remain in EMU settings.

5.3.4 Performance Metrics Reporting

Table 3 summarizes the heterogeneity in performance metric reporting, complicating cross-study comparison. Sensitivity ranges from 51.2% (forecasting mean) to 100% (tonic detection), while FPR reporting varies from per-hour to per-night, with several studies omitting this critical metric entirely.

Table 2 reveals additional clinically relevant metrics that are critical for real-world deployment but inconsistently reported:

- **Detection latency** is reported by only 4/13 studies, despite being critical for timely intervention.
- **Patient-level success rate** (proportion achieving usable performance) is reported by only 6/13 studies, ranging from 43.5% to 100%.
- **Precision/PPV** is rarely reported (3/13), limiting understanding of positive predictive value for clinical decision-making.

- **Generalization metrics** (LOSO validation, patient-independent testing) show that only Meisel et al. used true LOSO CV, with most studies reporting aggregated performance that may overestimate real-world utility.

Table 3: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/9	2/4	Reintjes, Fine, Spahr, Dong, Wang, Singh Rathore, Elemam, Borujeny, Meisel, Ode
Specificity	1/9	0/4	Elemam only
FPR/FA rate	7/9	2/4	Spahr, Reintjes, Fine, Dong, Wang, Ode, Meisel, Nasser, Stirling
AUC-ROC	1/9	2/4	Reintjes, Nasser, Stirling
Detection Latency	3/9	1/4	Spahr, Fine, Dong, Nasser
Patient Success Rate	2/9	4/4	Meisel, Nasser, Ode, Stirling, Vieluf, Wang
Precision/PPV	2/9	1/4	Ode, Singh Rathore, Elemam, Wang
IoC / Time in Warning	0/9	2/4	Meisel, Nasser

5.4 Synthesis and Gaps

The concept matrix and detailed metrics reveal several patterns:

1. **Clinical translation gap:** Despite high sensitivities (>90% in several detection studies), FPR remains clinically unacceptable for standalone deployment. Only Dong et al. (0.165/h) and Fine (0.023/h) achieve FARs approaching clinical targets (<0.1–0.5/h) (Beniczky & Ryvlin, 2018).
2. **Personalization vs. generalization:** LOSO approaches achieve lower but more realistic performance (51.2% sensitivity, 43.5% patient success) compared to patient-specific tuning. Meisel et al.’s LOSO CV results suggest that aggregated performance metrics may overestimate real-world utility.
3. **Patient-level heterogeneity:** Patient success rates vary dramatically (43.5–100%), with Meisel and Ode showing that fewer than half of patients achieve usable performance. Nasser (5/6) and Stirling (100%) demonstrate that responder identification remains a critical challenge.
4. **Diary integration:** (Vieluf et al., 2025) demonstrates that hybrid approaches (wearable + clinical data) may outperform wearables alone, suggesting a direction for ambulatory systems where wearable-only performance is insufficient.
5. **Metric heterogeneity:** Inconsistent reporting (sensitivity vs. accuracy vs. AUC vs. IoC) impedes meta-analysis and clinical benchmarking. Key clinical metrics like detection latency and patient success rate are reported by fewer than half of studies.

6 Key Findings and Clinical Takeaways

This section synthesizes the evidence from 13 primary studies and 5 exploratory feasibility studies on wearable seizure detection and forecasting, identifying patterns that inform clinical practice and future research directions.

6.1 Executive Summary

- **Current State:** Wearable seizure detection achieves high sensitivity (71–100%) for convulsive seizures using accelerometry, but false alarm rates remain clinically unacceptable in most studies.
- **Forecasting Challenge:** Seizure forecasting remains immature, with only 43.5–83% of patients achieving better-than-chance performance and significant heterogeneity in individual responses.
- **Best Performing Approaches:** Multimodal sensor fusion (ACC + cardiac) and ensemble deep learning methods show the most promise for clinical translation.
- **Validation Gap:** Only 3 of 13 studies include prospective validation, with most relying on retrospective analyses that may overestimate real-world performance.

6.2 Detection Studies: Performance Landscape

6.2.1 High-Sensitivity Detectors

Table 4 summarizes the best-performing detection systems across different seizure types and settings.

Table 4: Highest Performing Detection Studies by Metric

Study	Seizure Type	Sensitivity	FAR	Latency	Validation	Ready
Fine 2025	Tonic	100%	0.023/h	14 s	Phase 1	○
Dong 2026	Nocturnal severe	71.6%	0.165/h	<30 s	Prospective	●
Spahr 2025	Convulsive	96%	<1/8d	26 s	Phase 2	●
Reintjes 2025	All types	98.16%	1.91/h	–	Retro	○
Borujeny 2013	Motor	100%	0	–	3-pt	○

Key Finding: Only two studies (Dong, Fine) achieve false alarm rates approaching clinical targets (<0.5/h) while maintaining reasonable sensitivity. The Phase 2 Spahr study demonstrates that multi-center validation is feasible but still shows FAR above clinical targets.

6.2.2 Clinical Translation Barriers

1. **False Alarm Crisis:** Reintjes et al. report FAR of 1.91–39.75/h, 20–200× higher than clinically acceptable. This confirms that ECG-only approaches without motion artifact filtering are not viable for standalone deployment.
2. **Seizure Type Limitation:** High sensitivity (>90%) is consistently achieved only for convulsive/motor seizures. Non-motor seizures remain largely undetectable by current wearable approaches.
3. **Offline Processing:** Most algorithms (Fine, Wang, Singh Rathore) run on offline servers rather than embedded devices, limiting real-time applicability.

6.3 Forecasting Studies: The Prediction Challenge

6.3.1 Performance Spectrum

Forecasting performance varies dramatically across studies and patients (Table 5).

Table 5: Forecasting Study Performance Comparison

Study	Metric	Value	TiW	Patient Success	Lead Time
Meisel 2020	IoC	14.1%	43.7%	43.5% (30/69)	–
Nasseri 2021	AUC-ROC	0.75	0.9–7.2 h/d	83% (5/6)	33 s mean
Stirling 2021	AUC	0.74	–	100% (hourly)	Cyclic HR
Vieluf 2025	Sensitivity	82%	–	–	Daily
Ode 2023	Sensitivity	74%	–	44% (29/66)	–

Critical Finding: Only Meisel et al. used true LOSO cross-validation, revealing that patient-specific tuning inflates performance by 30–40 percentage points. Their 43.5% patient success rate suggests that fewer than half of patients may benefit from current forecasting approaches.

6.3.2 The Responder Problem

- Meisel (2020): 43.5% significant forecasters
- Ode (2023): 44% patient success (29/66 with 100% sensitivity)
- Nasseri (2021): 83% success (5/6 above chance)
- Stirling (2021): 100% hourly success but highly variable AUC

Interpretation: Patient heterogeneity is the fundamental challenge in seizure forecasting. Universal models (Vieluf) may report aggregated performance that masks significant individual variation.

6.4 Methodological Insights

6.4.1 Algorithm Evolution Trajectory

Table 6: Evolution of Algorithms in Seizure Detection/Forecasting (2013–2026)

Period	Algorithms	Representative Studies	Characteristics
2013–2015	KNN, ANN, SVM	Borujeny 2013, Fujiwara 2016	Handcrafted features
2016–2019	HRV-based MLP, LSTM	Yamakawa 2020, Billeci 2018	Patient-specific models
2020–2022	LSTM+1DConv, Ensembles	Meisel 2020, Stirling 2021	LOSOS cross-validation
2023–2026	CNN+Attention, Matrix Profile, Self-Att AE	Ode 2023, Reintjes 2025, Spahr 2025	Anomaly detection techniques

6.5 Exploratory Studies: Feasibility Evidence

6.5.1 Cardiac Signal Viability

The five exploratory studies collectively demonstrate that cardiac signals (ECG, HRV, PPG) contain detectable preictal changes:

- **Pavei 2017:** 94.1% sensitivity using SVM on HRV dynamics (n=12)
- **Fujiwara 2016:** 91% sensitivity using multivariate statistical process control
- **Leal 2017:** Identified RRMean and BPMMean as most consistent ECG features (n=167)
- **Jain 2017:** Demonstrated EDA decomposition feasibility for seizure-related autonomic changes
- **Seth 2023:** Systematic review showing 56–100% sensitivity range across 24 studies

Synthesis: Cardiac signals are viable for seizure detection but require sophisticated signal processing to overcome motion artifacts and individual variability.

6.6 Clinical Recommendations

6.6.1 For Current Clinical Practice

Based on current evidence, the following recommendations can be made:

1. **Nocturnal Monitoring:** Dong et al.’s NightWatch system (71.6% sensitivity, 0.165/h FAR) is the most validated option for home nocturnal monitoring.
2. **Tonic Seizures:** Fine et al.’s algorithm (100% sensitivity, 0.023/h FAR) shows promise for tonic seizure detection but requires Phase 2 validation.
3. **EMU Settings:** Spahr et al.’s Phase 2 validated algorithm (96% sensitivity, <1/8d FAR) is appropriate for EMU seizure counting.
4. **Avoid:** Standalone ECG-based detection systems (FAR >1/h) without confirmation from accelerometry.

6.6.2 For Researchers and Developers

1. **Validation Standard:** LOSO cross-validation should be the minimum standard for reporting performance.
2. **Metric Reporting:** All studies should report: sensitivity, specificity/PPV, FAR per hour, detection latency, and patient success rate.
3. **Interpretability:** SHAP or similar explainable AI frameworks should be incorporated to enable clinical trust and regulatory approval.
4. **Responder Identification:** Pre-study characterization of likely responders based on seizure type, HRV reactivity, and other biomarkers.

6.7 Limitations of Current Evidence

6.7.1 Study Design Limitations

- **Small Samples:** Median $n=15-70$ patients; most studies underpowered for subgroup analysis
- **Short Duration:** Median monitoring <1 week per patient; rare events (seizures) limit statistical power
- **Retrospective Bias:** 10/13 studies retrospective; high risk of data leakage and optimistic bias
- **Setting Bias:** Most studies in EMU; home performance likely worse

6.7.2 Metric Heterogeneity

- Sensitivity vs. accuracy vs. AUC reported inconsistently
- FPR reported as per-hour, per-day, per-night, or not at all
- Patient success rate reported by only 6/13 studies
- No standardized benchmark datasets (except EPILEPSIAE)

6.8 Future Directions

6.8.1 Research Priorities

Based on identified gaps, the following research priorities emerge:

1. **Standardized Metrics:** Develop consensus on minimum reporting standards (ILAE Phase-appropriate metrics)
2. **Responder Prediction:** Identify baseline biomarkers that predict which patients will achieve usable forecasting performance
3. **Edge Optimization:** More studies like Spahr (1.12 s inference) needed for real-time deployment
4. **Personalization at Scale:** Federated learning approaches that enable patient-specific models without data centralization
5. **Hybrid Systems:** Combined wearable + diary approaches (Vieluf) that leverage complementary information sources

6.8.2 Emerging Technologies

- **Matrix Profile Anomaly Detection:** Reintjes et al. show unsupervised approaches can achieve high sensitivity without seizure-specific training
- **Self-Attentive Autoencoders:** Ode et al. demonstrate AE approaches for RRI-based forecasting

- **Attention Mechanisms:** Dong et al. use attention to weight reliable signals and reduce motion artifact impact
- **Ensemble Deep Learning:** Spahr et al. show ensemble CNN approaches achieve Phase 2 validation

6.9 Conclusion

Wearable seizure detection and forecasting have advanced significantly from proof-of-concept studies to Phase 2 clinical validation. Detection of convulsive seizures is now reliable enough for limited clinical use, with two systems (Dong, Fine) achieving false alarm rates compatible with patient adoption. However, seizure forecasting remains an open challenge, with fewer than half of patients achieving better-than-chance performance in LOSO-validated studies.

The path forward requires: (1) rigorous LOSO validation as the standard, (2) standardized metric reporting enabling cross-study comparison, (3) interpretable models that earn clinician trust, and (4) recognition that patient heterogeneity may require personalized approaches rather than universal solutions. The integration of multimodal sensors, attention mechanisms, and explainable AI represents the current state-of-the-art, but clinical translation demands prospective, multi-center validation in real-world settings.

References

- AbuAlrob, M. A., Itbaisha, A., & Mesraoua, B. (2025). Unlocking new frontiers in epilepsy through AI: From seizure prediction to personalized medicine. *Epilepsy & Behavior*, 166, 110327. Retrieved November 21, 2025, from URL: <https://www.sciencedirect.com/science/article/pii/S1525505025000666> (cit. on p. 3).
- Beniczky, S., & Ryvlin, P. (2018). Standards for testing and clinical validation of seizure detection devices. *Epilepsia*, 59 Suppl 1, 9–13. DOI: [10.1111/epi.14049](https://doi.org/10.1111/epi.14049) (cit. on pp. 10, 12).
- Beniczky, S., Wiebe, S., Jeppesen, J., Tatum, W. O., Brazdil, M., Wang, Y., Herman, S. T., & Ryvlin, P. (2021). Automated seizure detection using wearable devices: A clinical practice guideline of the International League Against Epilepsy and the International Federation of Clinical Neurophysiology. *Clinical Neurophysiology*, 132(5), 1173–1184. DOI: [10.1016/j.clinph.2020.12.009](https://doi.org/10.1016/j.clinph.2020.12.009) (cit. on pp. 3, 4).
- Borujeny, G. T., Yazdi, M., Keshavarz-Haddad, A., & Borujeny, A. R. (2013). Detection of Epileptic Seizure Using Wireless Sensor Networks. *Journal of Medical Signals and Sensors*, 3(2), 63–68. Retrieved January 5, 2026, from URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3788195/> (cit. on p. 11).
- Chen, F., Chen, I., Zafar, M., Sinha, S. R., & Hu, X. (2022). Seizures detection using multimodal signals: A scoping review. *Physiological Measurement*, 43(7), 07TR01. DOI: [10.1088/1361-6579/ac7a8d](https://doi.org/10.1088/1361-6579/ac7a8d) (cit. on p. 3).
- Dong, C., Ye, T., Long, X., Aarts, R. M., van Dijk, J. P., Shang, C., Liao, X., Chen, W., Lai, W., Chen, L., & Wang, Y. (2022). A Two-Layer Ensemble Method for Detecting Epileptic Seizures Using a Self-Annotation Bracelet With Motor Sensors. *IEEE Transactions on Instrumentation and Measurement*, 71, 1–13. DOI: [10.1109/TIM.2022.3173270](https://doi.org/10.1109/TIM.2022.3173270) (cit. on p. 11).

- Elemam, M. E., Elgamal, A., Elmenshawi, I., & Abdelkader, H. E. (2025). Automated validated tool for epileptic seizure detection using deep learning. *Journal of Intelligent Systems and Internet of Things*, 15(1), 74–90. DOI: [10.54216/JISIoT.150107](https://doi.org/10.54216/JISIoT.150107) (cit. on p. 11).
- Ghaderi, F. (2025, April 11). *Advances in Machine Learning for Epileptic Seizure Prediction: A Review of ECG-Based Approaches*. 2025040942. DOI: [10.20944/preprints202504.0942.v1](https://doi.org/10.20944/preprints202504.0942.v1). (Cit. on p. 3).
- Hussein, A. F., Arunkumar, N., Gomes, C., Alzubaidi, A. K., Habash, Q. A., Santamaria-Granados, L., Mendoza-Moreno, J. F., & Ramirez-Gonzalez, G. (2018). Focal and Non-Focal Epilepsy Localization: A Review. *IEEE Access*, 6, 49306–49324. DOI: [10.1109/ACCESS.2018.2867078](https://doi.org/10.1109/ACCESS.2018.2867078) (cit. on p. 2).
- Jain, S., Oswal, U., Xu, K. S., Eriksson, B., & Haupt, J. (2017). A Compressed Sensing Based Decomposition of Electrodermal Activity Signals. *IEEE Transactions on Biomedical Engineering*, 64(9), 2142–2151. DOI: [10.1109/TBME.2016.2632523](https://doi.org/10.1109/TBME.2016.2632523) (cit. on p. 3).
- Kalousios, S., Müller, J., Yang, H., Eberlein, M., Uckermann, O., Schackert, G., Polanski, W. H., & Leonhardt, G. (2024). ECG-based epileptic seizure prediction: Challenges of current data-driven models. *Epilepsia Open*, 10(1), 143–154. DOI: [10.1002/epi4.13073](https://doi.org/10.1002/epi4.13073) (cit. on p. 4).
- Leal, A., da Graça Ruano, M., Henriques, J., de Carvalho, P., & Teixeira, C. (2017). On the viability of ECG features for seizure anticipation on long-term data. *2017 IEEE 3rd International Forum on Research and Technologies for Society and Industry (RTSI)*, 1–5. DOI: [10.1109/RTSI.2017.8065951](https://doi.org/10.1109/RTSI.2017.8065951) (cit. on p. 3).
- Lu, S., Liu, L., Li, J., Chambers, J., Cook, M. J., & Grayden, D. B. (2025). Leveraging Channel Coherence in Long-Term iEEG Data for Seizure Prediction. *IEEE Journal of Biomedical and Health Informatics*, 29(8), 5541–5548. DOI: [10.1109/JBHI.2025.3556775](https://doi.org/10.1109/JBHI.2025.3556775) (cit. on p. 4).
- Mason, F., Scarabello, A., Taruffi, L., Pasini, E., Calandra-Buonaura, G., Vignatelli, L., & Bisulli, F. (2024). Heart Rate Variability as a Tool for Seizure Prediction: A Scoping Review. *Journal of Clinical Medicine*, 13(3), 747. DOI: [10.3390/jcm13030747](https://doi.org/10.3390/jcm13030747) (cit. on p. 3).
- Meisel, C., El Atrache, R., Jackson, M., Schubach, S., Ufongene, C., & Loddenkemper, T. (2020). Machine learning from wristband sensor data for wearable, noninvasive seizure forecasting. *Epilepsia*, 61(12), 2653–2666. DOI: [10.1111/epi.16719](https://doi.org/10.1111/epi.16719) (cit. on p. 11).
- Miron, G., Halimeh, M., Jeppesen, J., Loddenkemper, T., & Meisel, C. (2025). Autonomic biosignals, seizure detection, and forecasting. *Epilepsia*, 66 Suppl 3, 25–38. DOI: [10.1111/epi.18034](https://doi.org/10.1111/epi.18034) (cit. on pp. 3, 4).
- Nasseri, M., Pal Attia, T., Joseph, B., Gregg, N. M., Nurse, E. S., Viana, P. F., Worrell, G., Dümpelmann, M., Richardson, M. P., Freestone, D. R., & Brinkmann, B. H. (2021). Ambulatory seizure forecasting with a wrist-worn device using long-short term memory deep learning. *Scientific Reports*, 11(1), 21935. DOI: [10.1038/s41598-021-01449-2](https://doi.org/10.1038/s41598-021-01449-2) (cit. on p. 11).
- Ode, R., Fujiwara, K., Miyajima, M., Yamakawa, T., Kano, M., Jin, K., Nakasato, N., Sawai, Y., Hoshida, T., Iwasaki, M., Murata, Y., Watanabe, S., Watanabe, Y., Suzuki, Y., Inaji, M., Kunii, N., Oshino, S., Khoo, H. M., Kishima, H., & Maehara, T. (2023). Development of an epileptic seizure prediction algorithm using R-R

- intervals with self-attentive autoencoder. *Artificial Life and Robotics*, 28(2), 403–409. DOI: [10.1007/s10015-022-00832-0](https://doi.org/10.1007/s10015-022-00832-0) (cit. on p. 11).
- Pavei, J., Heinzen, R. G., Novakova, B., Walz, R., Serra, A. J., Reuber, M., Ponnusamy, A., & Marques, J. L. B. (2017). Early Seizure Detection Based on Cardiac Autonomic Regulation Dynamics. *Frontiers in Physiology*, 8. DOI: [10.3389/fphys.2017.00765](https://doi.org/10.3389/fphys.2017.00765) (cit. on p. 3).
- Reintjes, C., Hagenbeck, J. F., Ballo, M., Rahlmeier, T., Wolf, S. M., & Schoder, D. (2025). ECG-Based Detection of Epileptic Seizures in Real-World Wearable Settings: Insights from the SeizeIT2 Dataset. *Sensors*, 25(24), 7687. DOI: [10.3390/s25247687](https://doi.org/10.3390/s25247687) (cit. on p. 11).
- Seth, E. A., Watterson, J., Xie, J., Arulsamy, A., Md Yusof, H. H., Ngadimon, I. W., Khoo, C. S., Kadirvelu, A., & Shaikh, M. F. (2023). Feasibility of cardiac-based seizure detection and prediction: A systematic review of non-invasive wearable sensor-based studies. *Epilepsia Open*, 9(1), 41–59. DOI: [10.1002/epi4.12854](https://doi.org/10.1002/epi4.12854) (cit. on p. 3).
- Shum, J., & Friedman, D. (2021). Commercially available seizure detection devices: A systematic review. *Journal of the Neurological Sciences*, 428, 117611. DOI: [10.1016/j.jns.2021.117611](https://doi.org/10.1016/j.jns.2021.117611) (cit. on p. 4).
- Sivathamboo, S., Nhu, D., Piccenna, L., Yang, A., Antonic-Baker, A., Vishwanath, S., Todaro, M., Yap, L. W., Kuhlmann, L., Cheng, W., O'Brien, T. J., Lannin, N. A., & Kwan, P. (2022). Preferences and User Experiences of Wearable Devices in Epilepsy: A Systematic Review and Mixed-Methods Synthesis. *Neurology*, 99(13), e1380–e1392. DOI: [10.1212/WNL.000000000000200794](https://doi.org/10.1212/WNL.000000000000200794) (cit. on p. 3).
- Spahr, A., Bernini, A., Ducouret, P., Baumgartner, C., Koren, J. P., Imbach, L., Beniczky, S., Larsen, S. A., Rheims, S., Fabricius, M., Seeck, M., Steinhoff, B. J., Beuchat, I., Dan, J., Atienza, D. A., Bardyn, C.-E., & Ryvlin, P. (2025). Deep learning-based detection of generalized convulsive seizures using a wrist-worn accelerometer. *Epilepsia*, 66 Suppl 3, 53–63. DOI: [10.1111/epi.18406](https://doi.org/10.1111/epi.18406) (cit. on p. 11).
- Stirling, R. E., Grayden, D. B., D'Souza, W., Cook, M. J., Nurse, E., Freestone, D. R., Payne, D. E., Brinkmann, B. H., Pal Attia, T., Viana, P. F., Richardson, M. P., & Karoly, P. J. (2021). Forecasting Seizure Likelihood With Wearable Technology. *Frontiers in Neurology*, 12, 704060. DOI: [10.3389/fneur.2021.704060](https://doi.org/10.3389/fneur.2021.704060) (cit. on p. 11).
- Sveinsson, O., Andersson, T., Mattsson, P., Carlsson, S., & Tomson, T. (2020). Clinical risk factors in SUDEP. *Neurology*, 94(4), e419–e429. DOI: [10.1212/WNL.00000000000008741](https://doi.org/10.1212/WNL.00000000000008741) (cit. on p. 2).
- Tugwell, P., & Tovey, D. (2021). PRISMA 2020. *Journal of Clinical Epidemiology*, 134, A5–A6. DOI: [10.1016/j.jclinepi.2021.04.008](https://doi.org/10.1016/j.jclinepi.2021.04.008) (cit. on p. 4).
- Vieluf, S., Tomioka, S., Zhang, B., Krishnan, V., Bosl, W. J., Grinnell, T., & Loddenkemper, T. (2025). Seizure monitoring by combined diary and wearable data: A multicenter, longitudinal, observational study. *Epilepsia*, 66(11), 4259–4271. DOI: [10.1111/epi.18550](https://doi.org/10.1111/epi.18550) (cit. on pp. 11, 12).
- Wang, J., Hu, D., Lai, X., Jiang, T., Jiang, T., Gao, F., Vidal, P.-P., & Cao, J. (2025). Epileptic Seizure Detection Based on Attitude Angle Signal of Wearable Device. *IEEE Transactions on Instrumentation and Measurement*, 74. DOI: [10.1109/TIM.2025.3529058](https://doi.org/10.1109/TIM.2025.3529058) (cit. on p. 3).
- Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS quarterly*, xiii–xxiii. Retrieved December 3, 2025, from URL: <https://www.jstor.org/stable/4132319> (cit. on pp. 4, 6).

- Wong, S., Simmons, A., Rivera-Villicana, J., Barnett, S., Sivathamboo, S., Perucca, P., Ge, Z., Kwan, P., Kuhlmann, L., Vasa, R., Mouzakis, K., & O'Brien, T. J. (2023). EEG datasets for seizure detection and prediction— A review. *Epilepsia Open*, 8(2), 252–267. DOI: [10.1002/epi4.12704](https://doi.org/10.1002/epi4.12704) (cit. on p. 4).
- Wu, D., Wei, J., Vidal, P.-P., Wang, D., Yuan, Y., Cao, J., & Jiang, T. (2024). A Novel Seizure Detection Method Based on the Feature Fusion of Multimodal Physiological Signals. *IEEE Internet of Things Journal*, 11(16), 27545–27556. DOI: [10.1109/JIOT.2024.3398418](https://doi.org/10.1109/JIOT.2024.3398418) (cit. on p. 3).